

DSA Pattern-Based Roadmap Checklist

Learn DSA through pattern recognition, invariants, and reusable mental models. Focus on understanding why patterns work instead of memorizing isolated problems.

1. Arrays & Strings

- ■ Traversal Pattern
 - ■ Prefix/Suffix Thinking
 - ■ Frequency Counting
 - ■ Simulation Problems
 - ■ Greedy Basics
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2. Two Pointers

- ■ Opposite Direction
 - ■ Read/Write Pointer
 - ■ Bidirectional Verification
 - ■ Sorted Array Elimination
 - ■ Controlled Mismatch
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3. Sliding Window

- ■ Fixed Size Window
 - ■ Variable Size Window
 - ■ HashSet Window
 - ■ Frequency Window
 - ■ Longest/Minimum Window
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4. Prefix Sum

- ■ Running Sum
 - ■ Subarray Sum Queries
 - ■ 2D Prefix Sum
 - ■ Difference Array
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5. HashMap / HashSet

- ■ Fast Lookup
 - ■ Frequency Counting
 - ■ Prefix Memory
 - ■ Window Support
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6. Stack

- ■ Balanced Parentheses
 - ■ Expression Evaluation
 - ■ Stack Simulation
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7. Monotonic Stack

- ■ Next Greater Element
 - ■ Next Smaller Element
 - ■ Previous Greater/Smaller
 - ■ Histogram
 - ■ Rain Water Trapping
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8. Queue / Deque

- ■ Basic Queue
 - ■ Circular Queue
 - ■ Deque
 - ■ Monotonic Queue
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9. Linked List

- ■ Traversal
- ■ Reversal
- ■ Slow/Fast Pointer
- ■ Cycle Detection

10. Recursion & Backtracking

- ■ Recursive Thinking
 - ■ Decision Tree
 - ■ Subsets
 - ■ Permutations
 - ■ Combination Problems
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11. Trees

- ■ DFS
 - ■ BFS
 - ■ Tree Recursion
 - ■ Diameter Problems
 - ■ BST
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12. Heap / Priority Queue

- ■ Top K Problems
 - ■ Merge K Structures
 - ■ Min/Max Heap
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13. Trie

- ■ Prefix Search
 - ■ Word Dictionary
 - ■ Autocomplete
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14. Graphs

- ■ BFS
- ■ DFS
- ■ Topological Sort

- ■ Union Find
 - ■ Shortest Path
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15. Greedy

- ■ Interval Scheduling
 - ■ Sorting + Greedy
 - ■ Local Optimal Choice
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16. Dynamic Programming

- ■ Memoization
 - ■ Tabulation
 - ■ 1D DP
 - ■ 2D DP
 - ■ State Transition
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17. Bit Manipulation

- ■ XOR Patterns
 - ■ Bitmasking
 - ■ Subset Enumeration
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18. Segment Tree / Fenwick Tree

- ■ Range Query
 - ■ Point Update
 - ■ Lazy Propagation
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19. String Algorithms

- ■ KMP
 - ■ Rabin-Karp
 - ■ Z Algorithm
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